This implies that with the same probability,

$$||\underline{\theta}^*_{\setminus u} - \underline{\hat{\theta}}_{\setminus u}||_\infty \leq \quad \sqrt{\varepsilon^2 + \frac{16\eta(1+\gamma)e^{4\gamma}}{\omega_P}} \leq \quad \varepsilon + 4e^{2\gamma}\sqrt{\frac{(1+\gamma)\eta}{\omega_P}} \tag{F17}$$

Now a union bound over each $u \in [n]$ gives us the desired bound in the theorem statement.

□

1. Technical lemmas for PL estimator without sparsity

a. *Gradient concentration*

Lemma 2. *For an η-strongly metastable $||\nabla\mathcal{L}(\underline{\theta}^*_u)||_\infty \leq \frac{4\eta}{\omega_P}$*

Proof.

$$\frac{\partial\mathcal{L}(\underline{\theta}^*_u)}{\partial\theta_{uk}} = \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \frac{-2\sigma_u\sigma_k}{1+\exp(2\sigma_u(\sum_{k'\neq u}\theta^*_{u,k'}\sigma_{k'}+\theta^*_u))}, \tag{F18}$$

$$= -2\mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mu(-\sigma_u|\sigma_{\setminus u})\sigma_u\sigma_k, \tag{F19}$$

$$= -2\mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \left(\mu(-\sigma_u|\sigma_{\setminus u}) - \nu(-\sigma_u|\sigma_{\setminus u})\right)\sigma_k\sigma_u. \tag{F20}$$

In the last line, we have used following relation that holds for all $k \neq u$,

$$\mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu}\nu(-\sigma_u|\sigma_{\setminus u})\sigma_u\sigma_k = \sum_{\underline{\sigma}}\nu(\underline{\sigma}_{\setminus u})\nu(\sigma_u|\underline{\sigma}_{\setminus u})\nu(-\sigma_u|\underline{\sigma}_{\setminus u})\sigma_u\sigma_k = \sum_{\underline{\sigma}_{\setminus u}}\nu(\underline{\sigma}_{\setminus u})\underbrace{\nu(\sigma_u|\underline{\sigma}_{\setminus u})\nu(-\sigma_u|\underline{\sigma}_{\setminus u})}_{\text{this is independent of }\sigma_u.}\sigma_k\sum_u\sigma_u = 0.$$

Now we can bound this gradient directly from Theorem 1,

$$\left|\frac{\partial\mathcal{L}(\underline{\theta}^*_u)}{\partial\theta_{u,k}}\right| \leq 2\mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} |\left(\mu(-\sigma_u|\sigma_{\setminus u}) - \nu(-\sigma_u|\sigma_{\setminus u})\right)| \tag{F21}$$

$$\leq \frac{4\eta}{\omega_P}. \tag{F22}$$

Similarly one can also show that,

$$\left|\frac{\partial\mathcal{L}(\underline{\theta}^*_u)}{\partial\theta_u}\right| \leq \frac{4\eta}{\omega_P}. \tag{F23}$$

□

Lemma 3. *For a $\eta-$strongly metastable distribution and $\varepsilon_a, \delta_a > 0$. Given $M \geq \frac{8}{\varepsilon_a^2}\log(\frac{2n}{\delta_a})$ i.i.d. samples guarantees with probability at least $1-\delta_a$ that,*

$$||\nabla\mathcal{L}_M(\underline{\theta}^*_u)||_\infty \leq \frac{4\eta}{\omega_P} + \varepsilon_a \tag{F24}$$

Proof. By direct calculation as done in the proof of Lemma 2,

$$\frac{\partial \mathcal{L}(\underline{\theta}_u^*)}{\partial \theta_{u,k}} = -2 \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mu(-\sigma_u|\sigma_{\setminus u})\sigma_u\sigma_k. \tag{F25}$$

Now the variable in the expectation can be easily bounded,

$$-1 \leq \sigma_u\sigma_k\mu(-\sigma_u|\underline{\sigma}_{\setminus u}) \leq 1. \tag{F26}$$

This implies from Hoeffding's inequality that,

$$Pr\left(\left|\frac{\partial \mathcal{L}(\underline{\theta}_u^*)}{\partial \theta_{u,k}} - \frac{\partial \mathcal{L}_M(\underline{\theta}_u^*)}{\partial \theta_{u,k}}\right| \geq \varepsilon_a\right) \leq 2\exp\left(\frac{-M\varepsilon_a^2}{8}\right). \tag{F27}$$

Repeating the same steps we can show that a similar tail bound also holds for the derivative w.r.t θ_u. Now choosing $M \geq \frac{8}{\varepsilon_a^2}\log(\frac{2n}{\delta_a})$ gives us the desired results by the union bound. □

b. Strong convexity-bounds on curvature

It is useful to define the following function on reals,

$$f(x) \coloneqq \log(1+\exp(-2x)). \tag{F28}$$

Lemma 4. *(Smoothnes and Strong convexity of f)*
Let f be as defined in (F28)*, and let $\delta f(x,\varepsilon) \coloneqq f(x+\varepsilon) - (f(x) + \varepsilon f'(x))$. Then if* $\max(|x|,|x+\varepsilon|) \leq \gamma$,

$$\frac{\varepsilon^2}{2} \geq \delta f(x,\varepsilon) \geq \frac{\exp(-2\gamma)\varepsilon^2}{2} \tag{F29}$$

Proof. This lemma can be shown using standard arguments from the strong convexity of f [60]. We can see that for any $|y| \leq \gamma$ we have,

$$f''(y) = \frac{2}{1+\cosh(2y)} \geq \frac{2}{1+\cosh(2\gamma)} \geq \exp(-2\gamma). \tag{F30}$$

From this, it is clear that f is convex in the domain $[-\gamma,\gamma]$. Moreover, the above expression gives that, $g(y) \coloneqq f(y) - \frac{exp(-2\gamma)}{2}y^2$ is also a convex function in the same domain. Now from the first-order convexity condition, $g(x+\varepsilon) \geq g(x) + \varepsilon g'(x)$, we can find the lower bound on δf.

Similarly, we can see that $f''(y) < 1$. This implies that $h(y) \coloneqq \frac{y^2}{2} - f(y)$ is also a convex function. From the relation $h(x+\varepsilon) \geq h(x) + \varepsilon h'(x)$ we get the upper bound on the δf. □

Lemma 5. $\delta\mathcal{L}(\Delta, \underline{\theta}_u^*) \geq \frac{e^{-4\gamma}}{2}||\Delta_{\setminus u}||_\infty^2 - \frac{8e^{-2\gamma}\eta}{\omega_P}$

Proof. Define local energy function $E_u(\underline{\sigma};\underline{\theta}) = \sum_{k\neq u}\theta_{u,k}\sigma_u\sigma_k + \theta_u\sigma_u$. Notice that this function is linear in the θ variables. Now PL loss can be expressed as a function of this local energy. From the definition of f in (F28), we have the following relations,

$$\mathcal{L}(\underline{\theta}_u + \Delta) = \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} f(E_u(\underline{\sigma};\underline{\theta}_u + \Delta)) = \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} f(E_u(\underline{\sigma};\underline{\theta}_u) + E_u(\underline{\sigma};\Delta)). \tag{F31}$$

$$\langle \nabla \mathcal{L}(\underline{\theta}_u), \Delta \rangle = \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} f'(E_u(\underline{\sigma}; \underline{\theta}_u))(\sum_{k \neq u} \frac{\partial E_u(\underline{\sigma}; \underline{\theta}_u)}{\partial \theta_{uk}} \Delta_{uk} + \frac{\partial E_u(\underline{\sigma}; \underline{\theta}_u)}{\partial \theta_u} \Delta_u) = \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} f'(E_u(\underline{\sigma}; \underline{\theta}_u)) E_u(\underline{\sigma}; \Delta). \tag{F32}$$

Now from the definitions in Lemma 4, we can define the curvature of $\mathcal{L}$ mimicking (F4) as follows,

$$\delta \mathcal{L}(\Delta, \underline{\theta}_u^*) = \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} \delta f(E_u(\underline{\sigma}, \underline{\theta}_u^*), E_u(\underline{\sigma}, \Delta)). \tag{F33}$$

Now from Condition 2 we have, $|E_u(\underline{\sigma}, \underline{\theta}_u^*)| \leq ||\underline{\theta}_u^*||_1 \leq \gamma$. The same condition will also hold for $|E_u(\underline{\sigma}, \hat{\underline{\theta}}_u)|_1$ as these constraints are imposed in the optimization.

This implies from Lemma 4 that,

$$\delta \mathcal{L}(\Delta, \underline{\theta}_u^*) \geq \frac{\exp(-2\gamma)}{2} \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} (E_u(\underline{\sigma}, \Delta))^2 = \frac{\exp(-2\gamma)}{2} \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} \left(\sum_{k \neq u} \sigma_k \Delta_{u,k} + \Delta_u \right) \geq \frac{\exp(-2\gamma)}{2} \mathop{\text{Var}}_{\underline{\sigma} \sim \nu} \left[\sum_{k \neq u} \Delta_{uk} \sigma_k \right]. \tag{F34}$$

Now for some $i \neq u$, we can use the law of conditional variances to bound this quantity. The choice of i will be made later to get optimal bounds.

$$\mathop{\text{Var}}_{\underline{\sigma} \sim \nu} \left[\sum_{k \neq u} \Delta_{uk} \sigma_k \right] \geq \mathop{\mathbb{E}}_{\underline{\sigma}_{\setminus i} \sim \nu_{\setminus i}} \mathop{\text{Var}}_{\sigma_i \sim \nu(.|\underline{\sigma}_{\setminus i})} \left(\sum_{k \neq u} \Delta_{uk} \sigma_k \middle| \underline{\sigma}_{\setminus i} \right) = \Delta_{ui}^2 \mathop{\mathbb{E}}_{\underline{\sigma}_{\setminus i} \sim \nu_{\setminus i}} \mathop{\text{Var}}_{\sigma_i \sim \nu(.|\underline{\sigma}_{\setminus i})} \left(\sigma_k \middle| \underline{\sigma}_{\setminus i} \right) \tag{F35}$$

Now as a consequence of the closeness of conditionals proved in Theorem 1, we can show that the conditional variance of the metastable state is close to that of the true distribution μ. This is shown in Lemma 7. Moreover the conditional variance w.r.t μ is naturally lower bounded by the finite temperature bound,

$$\mathop{\text{Var}}_{\sigma_u \sim \mu(.|\underline{\sigma}_{\setminus u})} [\sigma_u | \underline{\sigma}_{\setminus u}] = 1 - \tanh^2(\sum_{j \neq u} \theta_{uj}^* \sigma_j + \theta_u^*) \geq 1 - \tanh^2(\gamma) \geq e^{-2\gamma}. \tag{F36}$$

Now define a function that captures the error incurred in replacing the metastable distribution with the equilibrium distribution in the conditional variance,
$G(\underline{\sigma}_{\setminus i}) \coloneqq \mathop{\text{Var}}_{\sigma_i \sim \nu(.|\underline{\sigma}_{\setminus i})} \left[\sigma_i | \underline{\sigma}_{\setminus i} \right] - \mathop{\text{Var}}_{\sigma_i \sim \mu(.|\underline{\sigma}_{\setminus i})} \left[\sigma_i | \underline{\sigma}_{\setminus i} \right]$

Using this in (F35) we find,

$$\mathop{\text{Var}}_{\underline{\sigma} \sim \nu} \left[\sum_{k \neq u} \Delta_{uk} \sigma_k \right] \geq \Delta_{ui}^2 \left(e^{-2\gamma} - \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} |G(\underline{\sigma}_{\setminus i})| \right) \tag{F37}$$

(F38)

Now by using the result in Lemma 7 we can upper bound $\mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} |G(\underline{\sigma}_{\setminus i})|$. This gives us

$$\mathop{\text{Var}}_{\underline{\sigma} \sim \nu} \left[\sum_{k \neq u} \Delta_{uk} \sigma_k \right] \geq \Delta_{ui}^2 \left(e^{-2\gamma} - \frac{4\eta}{\omega_P} \right) = ||\Delta_{\setminus u}||_\infty^2 \left(e^{-2\gamma} - \frac{4\eta}{\omega_P} \right) \tag{F39}$$

(F40)

We have chosen the i here such that $\Delta_{ui}^2 = ||\Delta_{\setminus u}||_\infty^2$. Plugging this back in (F34) gives us the desired result.

□